

YASH PANDEY

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CSE - Shift - I (2nd Year - 2023)

MAIT ROHINI

OOPs USING C++ (2023-24) - 3rd Sem
(Complete OOPs Lab File)

Object-Oriented Programming Using C++ Lab
(CIC-257)

Faculty Name : Ms. Vaishali

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Semester : 3rd Semester



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MAHARAJA AGRASEN INSTITUTE OF TECHNOLOGY

Computer Science & Engineering Department

VISION

"To be centre of excellence in education, research and technology transfer in the field of computer engineering and promote entrepreneurship and ethical values."

MISSION

To foster an open, multidisciplinary and highly collaborative research environment for producing world-class engineers capable of providing innovative solutions to real-life problems and fulfil societal needs.

Lab Assessment Sheet

S.No	Experiment	Marks					Total Marks	Date	Signature
		R1	R2	R2	R4	R5			
<u>1.</u>	WAP To Print Hello World							31-8-23	Excellent performance
<u>2.</u>	WAP To Print Number from 1 To 10	3	3	3	3	3	15	31-8-23	
<u>3.</u>	WAP for Menu driven program for Matrices							31-8-23	14/09
<u>4.</u>	WAP For Class Student & Member F ⁿ of Class Student							14-9-23	Excellent performance
<u>5.</u>	WAP To Implement Function Overloading	3	3	3	3	3	15	14-9-23	Excellent performance
<u>6.</u>	WAP To Use Class For Conversion of Time in H, M, S							14-4-23	21/09
<u>7.</u>	WAP for Class Student & Member F ⁿ of Class Student	3	3	3	3	3		21-9-23	Excellent work
<u>8.</u>	Program To Generate Fibonacci series Using Constructor						15	21-9-23	
<u>9.</u>	WAP To Find Sum of Complex N.O. using Constructor Overloading							21-9-23	05/10
<u>10.</u>	WAP for Greatest of 2 No. Using Class & friend F ⁿ	3	3	3	3	3		5-10-23	

S.No	Experiment	Marks					Total Marks	Date	Signature
		R1	R2	R2	R4	R5			
<u>11.</u>	WAP To find Great of 3 No. using class and friend function							5-10-23	Excellent. ① good 12/10/23
<u>12.</u>	WAP To find Sum of 2 No. using Class & Friend FN							5-10-23	
<u>13.</u>	WAP To Implement String and perform operator overloading	3	3	3	3	3	(15)	12-10-23	② good ① good 16/11/23
	like (+), (==) and (<=) using operator overloading								
<u>14.</u>	WAP and Implement String & perform FN on Strings							12-10-23	Excellent
<u>15.</u>	WAP for overloading of new & delete operator	3	3	3	3	3	(15)	19-10-23	
<u>16.</u>	WAP To overload Unary (++) Operator							19-10-23	① good 16/11/23
<u>17.</u>	WAP To Implement Single Inheritance in C++							19-10-23	
<u>18.</u>	WAP To Implement Multiple Inheritance in C++ language	3	3	3	3	3	(15)	26-10-23	Excellent
<u>19.</u>	WAP To Implement Multilevel Inheritance in C++ language							26-10-23	

S.No	Experiment	Marks					Total Marks	Date	Signature
		R1	R2	R2	R4	R5			
<u>20.</u>	WAP To Implement Virtual F ⁿ with classes in C++							26-10-23	① gnel 18/11/23
<u>21.</u>	WAP To Implement Pure Virtual F ⁿ with classes	3	3	3	3	3	(15)	16-11-23	① gnel Excellent 30/11/23
<u>22.</u>	WAP of Function Template To find Square of a No.	3	3	3	3	3	(15)	30-11-23	① gnel Excellent
<u>23.</u>	WAP To perform Special F ⁿ , Constructors and destructors							30-11-23	① gnel 5/12/23
	in The class — Template with help of F ⁿ Template								
<u>24.</u>	WAP To Change Spaces (all types) with File Handling							30-11-23	
<u>25.</u>	WAP To Input data from The user and store in an	3	3	3	3	3	(15)	30-11-23	① gnel Excellent
	Text file Using File Handling & Then Display Them.								① gnel 5/12/23
<u>26.</u>	WAP To Raise an Exception if any attempts are made							30-11-23	
	To Refer an Element outside The Size Of The Array.								

Lab Assessment Sheet

[illegible]

S.No	Experiment	Marks					Total Marks	Date	Signature
		R1	R2	R2	R4	R5			

PROGRAM-1

Q :- Write A Program To Print Hello World As Output ?

Sol :- #include<iostream>
using namespace std;

```
int main(){  
    cout<<"HELLO WORLD"<<endl;  
    return 0;  
}
```



The screenshot shows a terminal window with the following content:

```
PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL  
PS C:\Users\ISHA PANDEY\Coding Isha> cd "c:\Users\ISHA PANDEY\Coding Isha\.vscode\" ; if ($?) { g++ yash.cpp -o yash } ; if ($?) { .\yash }  
HELLO WORLD  
PS C:\Users\ISHA PANDEY\Coding Isha\.vscode>
```

PROGRAM-2

Q :- WAP To Print Number 1-10 Using For Loop?

Sol :- `#include<iostream>`

`using namespace std;`

`int main(){`

`for(int i=1 ; i<=10 ; i++){`

`cout<<i<<" ";`

`}`

`return 0;`

`}`



```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL
PS C:\Users\ISHA PANDEY\Coding Isha> cd "c:\Users\ISHA PANDEY\Coding Isha\.vscode\" ; if ($?) { g++ yash.cpp -o yash } ; if ($?) { .\yash }
1 2 3 4 5 6 7 8 9 10
PS C:\Users\ISHA PANDEY\Coding Isha\.vscode>
```

PROGRAM-3

Q :- WAP To Menu-Driven A Program For Add, Subtract And Multiply Of Matrix Using OOPs?

Sol :-

```
#include<iostream>
```

```
using namespace std;
```

```
void Addition(int arr1[2][2] , int arr2[2][2]){
```

```
    int arr[2][2];
```

```
    cout<<"PERFORMING ADD OPERATION ON THE 2 MATRIX....."<<endl;
```

```
    for(int i=0 ; i<2 ; i++){
```

```
        for(int j=0 ; j<2 ; j++){
```

```
            arr[i][j] = arr1[i][j] + arr2[i][j];
```

```
            cout<<"THE VALUE AT arr["<<i<<"]["<<j<<"] IS ::: "<<arr[i][j]<<endl;
```

```
        }
```

```
    }
```

```
    cout<<endl;
```

```
}
```

```

void Subtract(int arr1[2][2] , int arr2[2][2]){
    int arr[2][2];
    cout<<"PERFORMING SUBTRACT OPERATION ON THE 2 MATRIX"<<endl;
    for(int i=0 ; i<2 ; i++){
        for(int j=0 ; j<2 ; j++){
            arr[i][j] = arr1[i][j] - arr2[i][j];
            cout<<"THE VALUE AT arr["<<i<<"]["<<j<<"] IS ::: "<<arr[i][j]<<endl;
        }
    }
    cout<<endl;
}

```

```

void Multiply(int arr1[2][2] , int arr2[2][2]){
    int arr[2][2];
    cout<<"PERFORMING MULTIPLY OPERATION ON THE 2 MATRIX"<<endl;
    for(int i=0 ; i<2 ; i++){
        for(int j=0 ; j<2 ; j++){
            arr[i][j] = 0;
            for(int k =0 ; k<2 ; k++){
                arr[i][j] += ( arr1[i][k] * arr2[k][j] );
            }
            cout<<"THE VALUE AT arr["<<i<<"]["<<j<<"] IS ::: "<<arr[i][j]<<endl;
        }
    }
    cout<<endl;
}

```

```

int main(){

    int arr1[2][2] = {{1,2} , {3,4}};

    int arr2[2][2] = {{4,3} , {2,1}};

    Addition(arr1 , arr2);    //Calling Addition Function.....

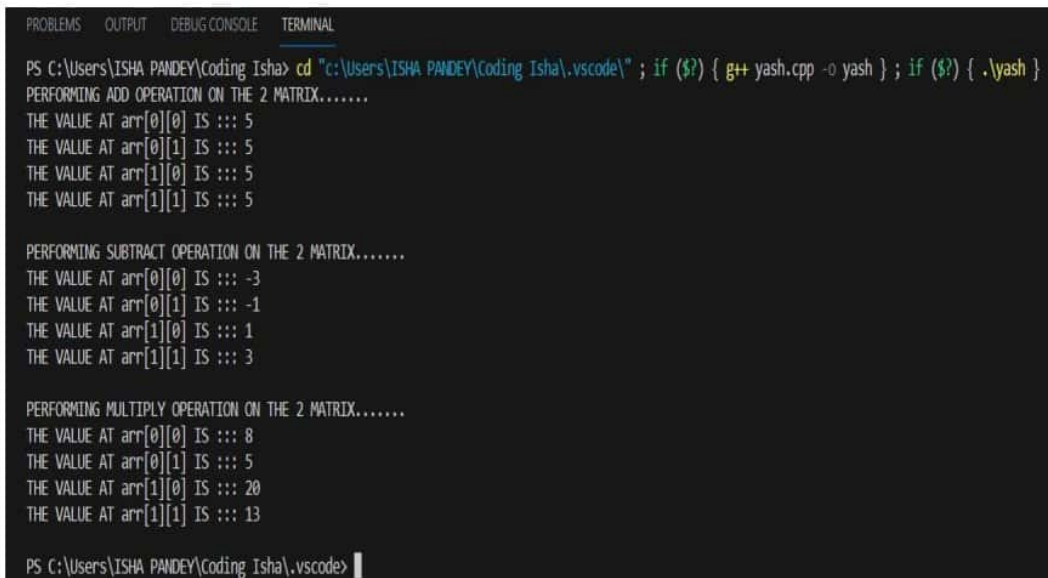
    Subtract(arr1 , arr2);    //Calling Subtract Function.....

    Multiply(arr1 , arr2);    //Calling Multiply Function.....

    return 0;

}

```



```

PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL
PS C:\Users\ISHA PANDEY\Coding Isha> cd "c:\Users\ISHA PANDEY\Coding Isha\.vscode\" ; if ($?) { g++ yash.cpp -o yash } ; if ($?) { .\yash }
PERFORMING ADD OPERATION ON THE 2 MATRIX.....
THE VALUE AT arr[0][0] IS ::: 5
THE VALUE AT arr[0][1] IS ::: 5
THE VALUE AT arr[1][0] IS ::: 5
THE VALUE AT arr[1][1] IS ::: 5

PERFORMING SUBTRACT OPERATION ON THE 2 MATRIX.....
THE VALUE AT arr[0][0] IS ::: -3
THE VALUE AT arr[0][1] IS ::: -1
THE VALUE AT arr[1][0] IS ::: 1
THE VALUE AT arr[1][1] IS ::: 3

PERFORMING MULTIPLY OPERATION ON THE 2 MATRIX.....
THE VALUE AT arr[0][0] IS ::: 8
THE VALUE AT arr[0][1] IS ::: 5
THE VALUE AT arr[1][0] IS ::: 20
THE VALUE AT arr[1][1] IS ::: 13

PS C:\Users\ISHA PANDEY\Coding Isha\.vscode>

```


PROGRAM-4

Q :- WAP To Create A Class Student In The Program Which Have Data-Members As Name , Branch , RollNo , Age And Marks Of 5 Subjects. Display The Name Of Those Students & % Who Got More Than 70% Marks?

Sol :-

```
#include<iostream>
```

```
using namespace std;
```

```
class Student{
```

```
    public:
```

```
        char Name[30];
```

```
        char Branch[30];
```

```
        int Age;
```

```
        int SubMarks[5];
```

```
        int RollNo;
```

```
        Student(int RNo,int age){
```

```
            this->RollNo=RNo;
```

```
            this->Age=age;
```

```
            cout<<"ENTER THE NAME OF THE STUDENT ::: ";
```

```
            cin>>Name;
```

```

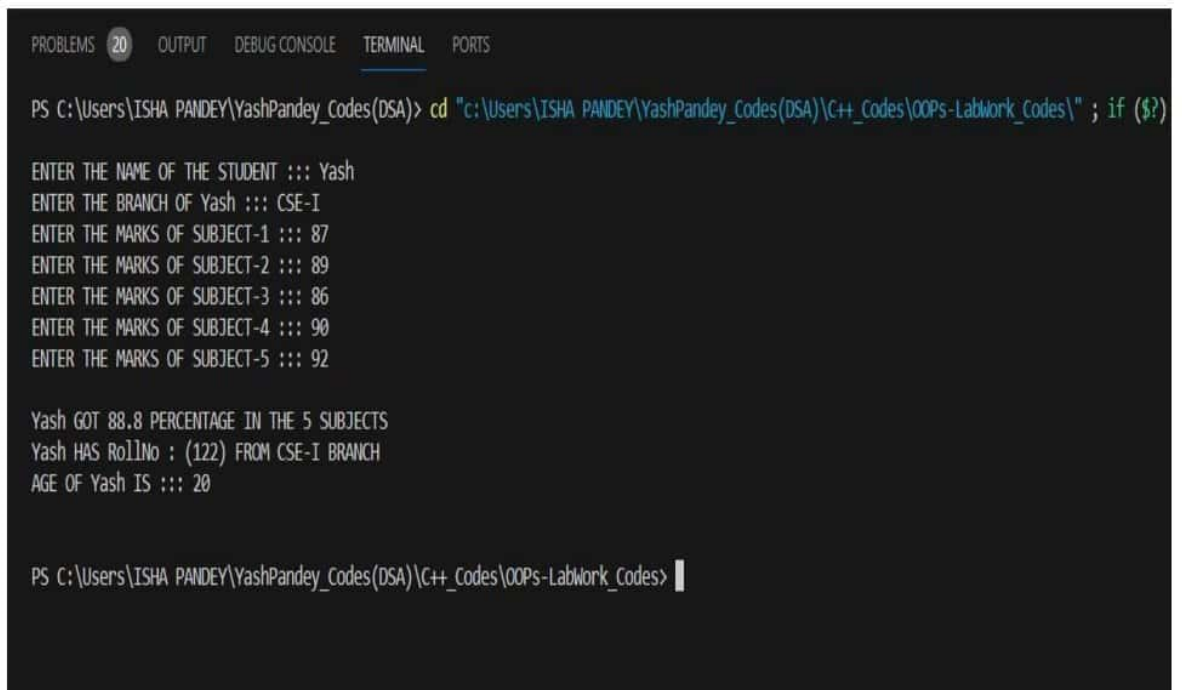
        cout<<"ENTER THE BRANCH OF "<<Name<<" :: ";
        cin>>Branch;
        for(int i=0 ; i < 5 ; i++){
            cout<<"ENTER THE MARKS OF SUBJECT-"<<(i+1)<<" :: ";
            cin>>SubMarks[i];
        }
        cout<<endl;
    }

    void MarksGrade( ){
        float TotalMarks=0.0;
        for(int i=0 ; i < 5 ; i++){
            TotalMarks += SubMarks[i];
        }
        float MarksAvg=TotalMarks/5.0;
        if(MarksAvg > 70.0){
            cout<<Name<<" GOT "<<MarksAvg<<" PERCENTAGE IN THE 5
                SUBJECTS"<<endl;
            cout<<Name<<" HAS RollNo : ("<<RollNo<<") FROM
                "<<Branch<<" BRANCH"<<endl;
            cout<<"AGE OF "<<Name<<" IS :: "<<Age<<endl;
        }
    }
};

```

```
int main(){  
    Student Yash(122,20);  
    Yash.MarksGrade();  
    return 0;  
}
```

OUTPUT : -



```
PROBLEMS 20 OUTPUT DEBUG CONSOLE TERMINAL PORTS  
PS C:\Users\ISHA PANDEY\YashPandey_Codes(DSA)> cd "c:\Users\ISHA PANDEY\YashPandey_Codes(DSA)\C++_Codes\OOPS-LabWork_Codes\" ; if ($?)  
  
ENTER THE NAME OF THE STUDENT ::: Yash  
ENTER THE BRANCH OF Yash ::: CSE-I  
ENTER THE MARKS OF SUBJECT-1 ::: 87  
ENTER THE MARKS OF SUBJECT-2 ::: 89  
ENTER THE MARKS OF SUBJECT-3 ::: 86  
ENTER THE MARKS OF SUBJECT-4 ::: 90  
ENTER THE MARKS OF SUBJECT-5 ::: 92  
  
Yash GOT 88.8 PERCENTAGE IN THE 5 SUBJECTS  
Yash HAS RollNo : (122) FROM CSE-I BRANCH  
AGE OF Yash IS ::: 20  
  
PS C:\Users\ISHA PANDEY\YashPandey_Codes(DSA)\C++_Codes\OOPS-LabWork_Codes>
```

PROGRAM-5.

Q :- WAP With Help Of Function Overloading To Calculate The Area Of Rectangle , Circle & Triangle ?

Sol :-

```
#include <iostream>
```

```
using namespace std;
```

```
// Function To Calculate Area Of Circle .....
```

```
float Area(float Radius){
```

```
    return (3.14) * Radius * Radius;
```

```
}
```

```
// Function To Calculate Area Of Rectangle .....
```

```
int Area(int length, int breath){
```

```
    return length * breath;
```

```
}
```

```
//Function To Calculate Area Of Triangle.....
```

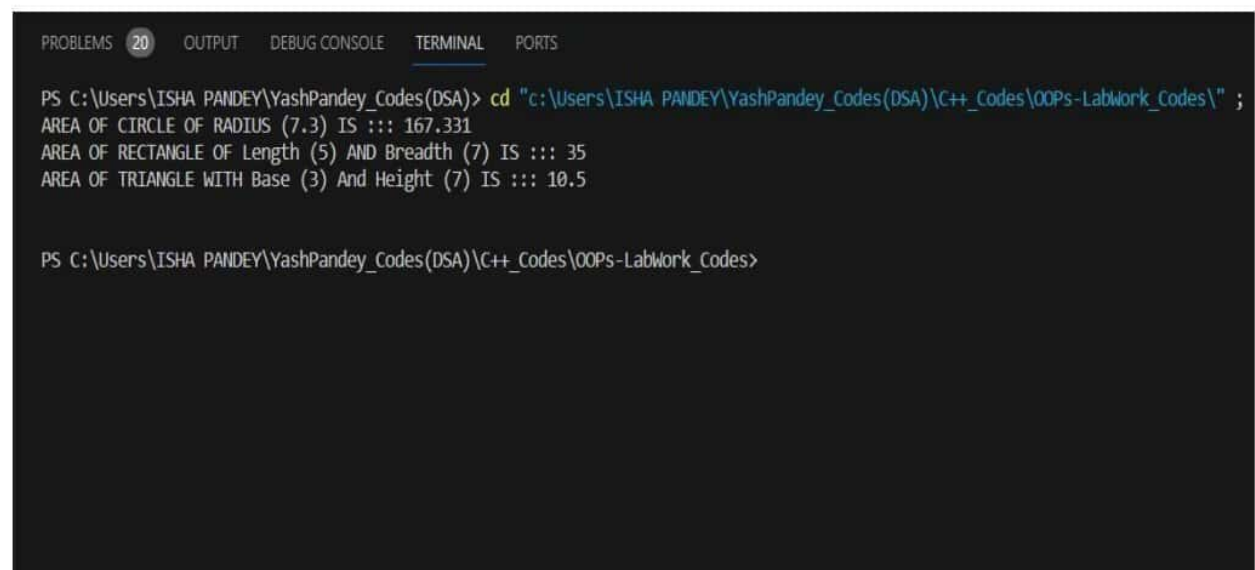
```
float Area(float b, float h){
```

```
    return (0.5) * b * h;
```

```
}
```

```
int main(){  
    float radius=7.3;  
    int Len=5,Bth=7;  
    float B=3,H=7;  
    cout<<"AREA OF CIRCLE OF RADIUS ("<<radius<<") IS :::  
    "<<Area(radius)<<endl;  
    cout<<"AREA OF RECTANGLE OF Length ("<<Len<<") AND Breadth  
    ("<<Bth<<") IS ::: "<<Area(Len,Bth)<<endl;  
    cout<<"AREA OF TRIANGLE WITH Base ("<<B<<") And Height ("<<H<<")  
    IS ::: "<<Area(B,H)<<endl;  
    return 0;  
}
```

OUTPUT : -



```
PROBLEMS 20 OUTPUT DEBUG CONSOLE TERMINAL PORTS  
PS C:\Users\ISHA PANDEY\YashPandey_Codes(DSA)> cd "c:\Users\ISHA PANDEY\YashPandey_Codes(DSA)\C++_Codes\OOPs-LabWork_Codes\" ;  
AREA OF CIRCLE OF RADIUS (7.3) IS ::: 167.331  
AREA OF RECTANGLE OF Length (5) AND Breadth (7) IS ::: 35  
AREA OF TRIANGLE WITH Base (3) And Height (7) IS ::: 10.5  
  
PS C:\Users\ISHA PANDEY\YashPandey_Codes(DSA)\C++_Codes\OOPs-LabWork_Codes>
```

PROGRAM-6.

Q :- WAP Which Has Class 'Time' With Data Members As Hour , Minute & Second . Take Input For 2 Objects For Class Function & Display The Result As Sum?

Sol :-

```
#include <iostream>
```

```
using namespace std;
```

```
class Time{
```

```
    int Hours;
```

```
    int Minutes;
```

```
    int Seconds;
```

```
    long TotalTime;
```

```
    public :
```

```
        Time(long tt){
```

```
            TotalTime=tt;
```

```
            Seconds=TotalTime;
```

```
            Minutes=(TotalTime/60);
```

```
            Seconds=TotalTime-(Minutes*60);
```

```
            Hours=Minutes/60;
```

```
            Minutes=Minutes-(Hours*60);
```

```
        }
```



```

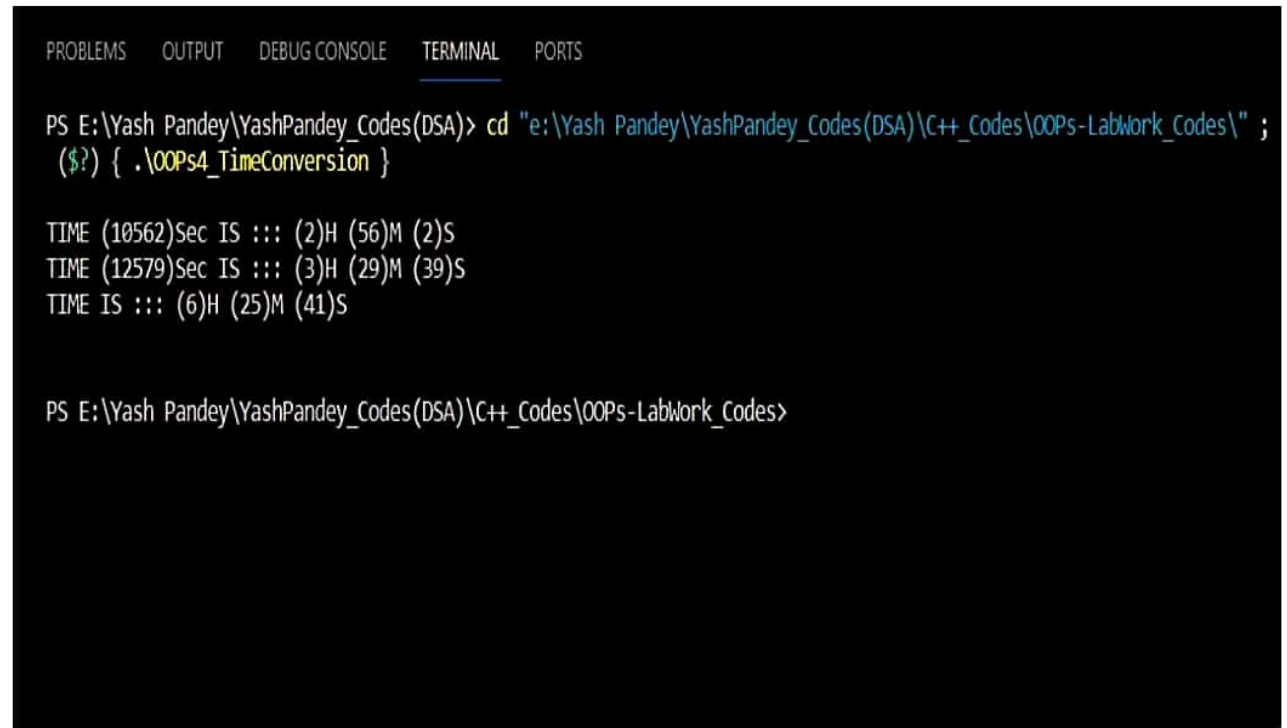
void DisplayTime(){
    cout<<"TIME ("<<TotalTime<<")Sec IS ::: ("<<Hours<<")H
        ("<<Minutes<<")M ("<<Seconds<<")S"<<endl;
}

void TimeSum(Time t1,Time t2){
    Seconds=t1.Seconds+t2.Seconds;
    int min=Seconds/60;
    Seconds=Seconds%60;
    Minutes=min+t1.Minutes+t2.Minutes;
    int hrs=Minutes/60;
    Minutes=Minutes%60;
    Hours=hrs+t1.Hours+t2.Hours;
    cout<<"TIME IS ::: ("<<Hours<<")H ("<<Minutes<<")M
        ("<<Seconds<<")S"<<endl;
}
};

int main(){
    Time t1(10562);
    t1.DisplayTime();
    Time t2(12579);
    t2.DisplayTime();
    Time t3;
    t3.TimeSum(t1,t2);
    return 0;
}

```

OUTPUT : -



The screenshot shows a terminal window with a dark background and light-colored text. At the top, there are tabs labeled 'PROBLEMS', 'OUTPUT', 'DEBUG CONSOLE', 'TERMINAL' (which is active and underlined), and 'PORTS'. The terminal content shows a command prompt session. The user is in a directory 'E:\Yash Pandey\YashPandey_Codes(DSA)' and runs 'cd "e:\Yash Pandey\YashPandey_Codes(DSA)\C++_Codes\OOPs-LabWork_Codes\" ; (\$?) { .\OOPs4_TimeConversion }'. The program then outputs three lines of time: 'TIME (10562)Sec IS ::: (2)H (56)M (2)S', 'TIME (12579)Sec IS ::: (3)H (29)M (39)S', and 'TIME IS ::: (6)H (25)M (41)S'. Finally, the prompt returns to 'PS E:\Yash Pandey\YashPandey_Codes(DSA)\C++_Codes\OOPs-LabWork_Codes>'.

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

PS E:\Yash Pandey\YashPandey_Codes(DSA)> cd "e:\Yash Pandey\YashPandey_Codes(DSA)\C++_Codes\OOPs-LabWork_Codes\" ;
($?) { .\OOPs4_TimeConversion }

TIME (10562)Sec IS ::: (2)H (56)M (2)S
TIME (12579)Sec IS ::: (3)H (29)M (39)S
TIME IS ::: (6)H (25)M (41)S

PS E:\Yash Pandey\YashPandey_Codes(DSA)\C++_Codes\OOPs-LabWork_Codes>
```

PROGRAM-7.

Q :- WAP Which Has Class Student And Members As Name , Marks of 5 Subjects . Display Name And ‘%’ Of Those Students Who Got Greater Than 70% Marks?

Sol :-

```
#include<iostream>

using namespace std;

class Student{
    private:
        char Name[30];
        float total;
    public:
        Student(int n1 , int n2 , int n3 , int n4 , int n5){
            cout<<"ENTER THE NAME OF THE STUDENT ::: ";
            cin>>Name;
            total=n1+n2+n3+n4+n5;
            cout<<endl;
        }
        void AvgMarks(){
            float avg=total/5.0;
```

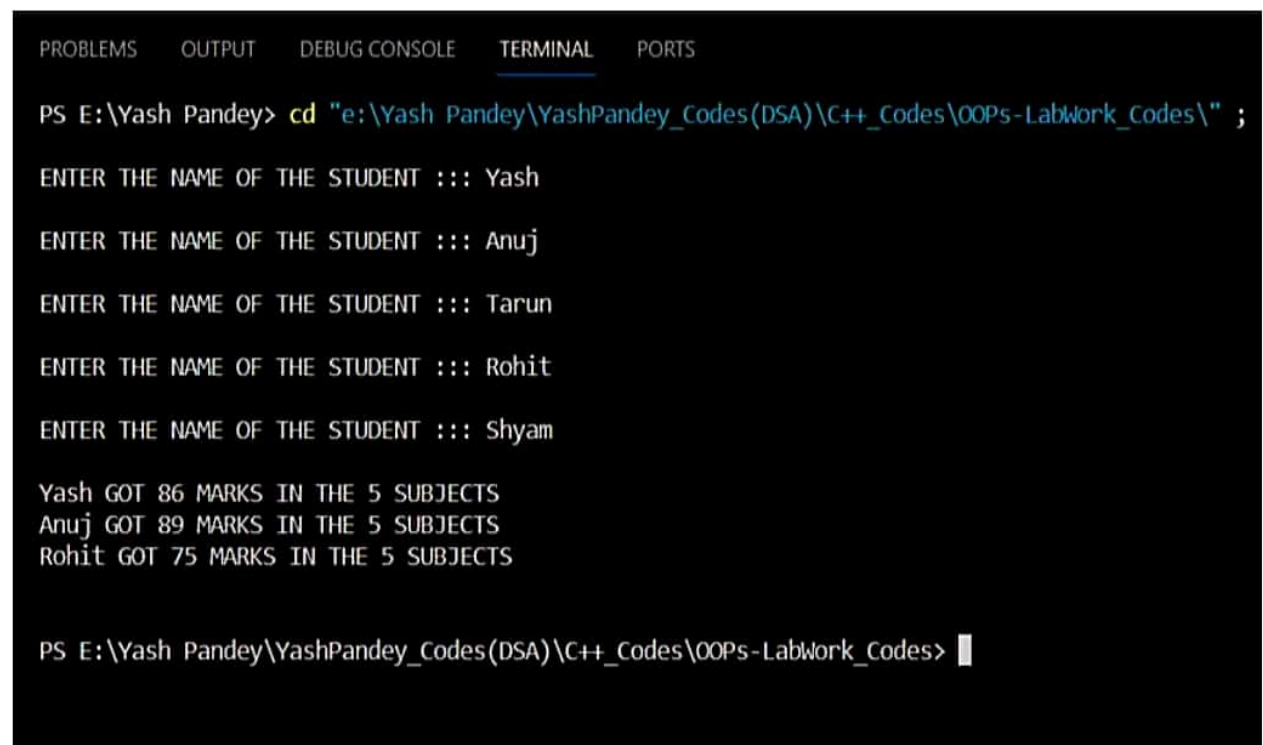
```

        if(avg>70.0){
            cout<<Name<<" GOT "<<avg<<" MARKS IN THE SUBJECTS\n";
        }
    }
};

int main(){
    Student s1(87,89,90,84,80) , s2(97,89,82,84,93) , s3(57,59,60,64,70);
    Student s4(67,79,75,84,70) , s5(71,67,65,62,76);
    s1.AvgMarks(); s2.AvgMarks(); s3.AvgMarks();
    s4.AvgMarks(); s5.AvgMarks();
    return 0;
}

```

OUTPUT : -



The screenshot shows a terminal window with a dark background. At the top, there are tabs for 'PROBLEMS', 'OUTPUT', 'DEBUG CONSOLE', 'TERMINAL' (which is active), and 'PORTS'. The terminal shows the following sequence of commands and outputs:

```

PS E:\Yash Pandey> cd "e:\Yash Pandey\YashPandey_Codes(DSA)\C++_Codes\OOPs-Labwork_Codes\" ;
ENTER THE NAME OF THE STUDENT ::: Yash
ENTER THE NAME OF THE STUDENT ::: Anuj
ENTER THE NAME OF THE STUDENT ::: Tarun
ENTER THE NAME OF THE STUDENT ::: Rohit
ENTER THE NAME OF THE STUDENT ::: Shyam

Yash GOT 86 MARKS IN THE 5 SUBJECTS
Anuj GOT 89 MARKS IN THE 5 SUBJECTS
Rohit GOT 75 MARKS IN THE 5 SUBJECTS

PS E:\Yash Pandey\YashPandey_Codes(DSA)\C++_Codes\OOPs-LabWork_Codes>

```

PROGRAM-8.

Q :- WAP To Generate Fibonacci Series Using Copy Constructor In The Class 'Fibonacci' ?

Sol :-

```
#include <iostream>

using namespace std;

class fibonacci{
    public:
        int val;
        fibonacci(){ val = 0; }
        fibonacci(fibonacci &obj){
            cout << "Calling The Copy Constructor Of Class Fibonacci...\n";
            val = obj.val;
        }
        int value(int n){
            if(n == 0){ return 0; }
            else if(n == 1){ return 1; }
            return value(n-1) + value(n-2);
        }
}
```

```

        void setValue(int n){ val = value(n); }

};

int main(){

    int index;

    cout << "ENTER THE INDEX FOR FIBONACCIONACCI SERIES ::: ";

    cin >> index;

    fibonacci f1;

    f1.setValue(index);

    fibonacci f2(f1);

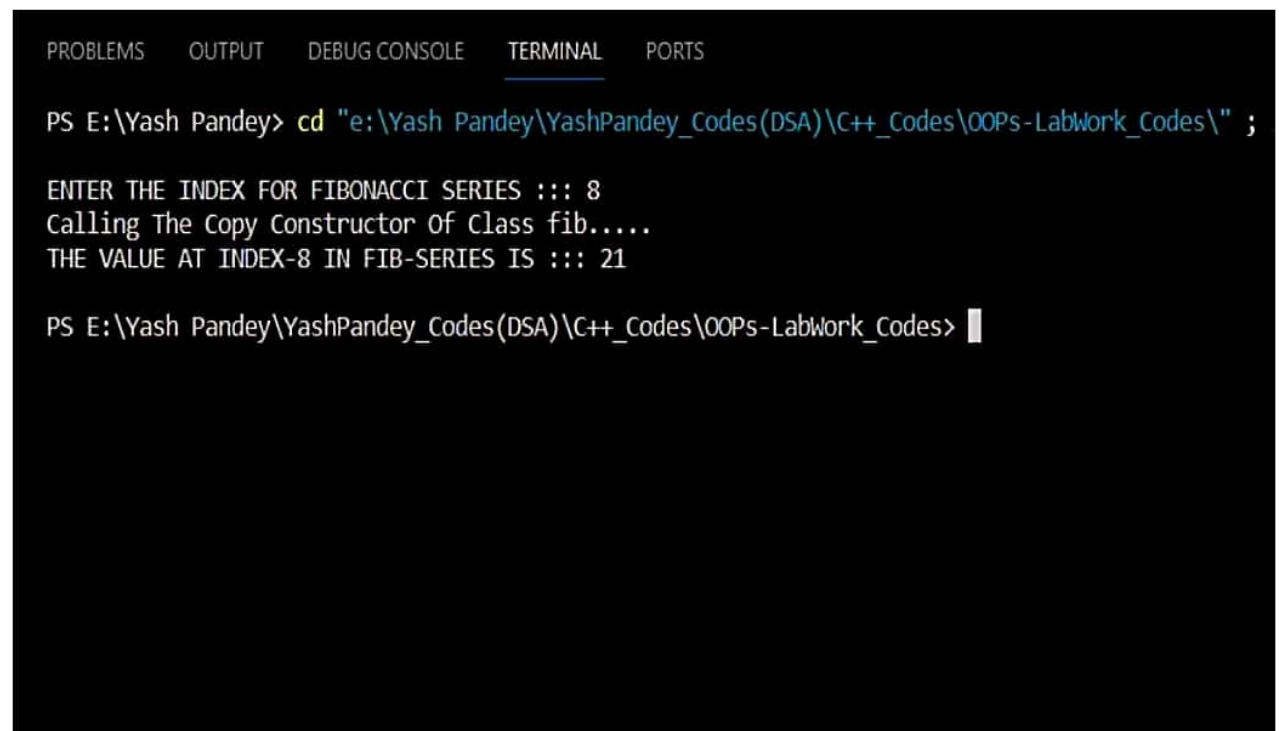
    cout << "THE VALUE AT INDEX-"<<index<<" IN FIB-SERIES IS ::: "<< f2.val;

    return 0;

}

```

OUTPUT : -



The screenshot shows a terminal window with a dark background and light-colored text. At the top, there are tabs for 'PROBLEMS', 'OUTPUT', 'DEBUG CONSOLE', 'TERMINAL' (which is active), and 'PORTS'. The terminal content shows the execution of a C++ program. It starts with a PowerShell prompt where the user navigates to a directory. Then, the program prompts for an index, which is 8. It then prints the value at that index, which is 21. The prompt ends with a cursor.

```

PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

PS E:\Yash Pandey> cd "e:\Yash Pandey\YashPandey_Codes(DSA)\C++_Codes\OOPs-LabWork_Codes\" ;

ENTER THE INDEX FOR FIBONACCI SERIES ::: 8
Calling The Copy Constructor Of Class fib.....
THE VALUE AT INDEX-8 IN FIB-SERIES IS ::: 21

PS E:\Yash Pandey\YashPandey_Codes(DSA)\C++_Codes\OOPs-LabWork_Codes> █

```


PROGRAM-9.

Q :- WAP To Perform Addition of Two Complex Numbers using Constructor Overloading Where The 1st Constructor Which Takes No Argument & Used To Create Objects Which Are Not Initialized , 2nd Constructor Takes One Argument & Is Used To Initialize Real Part & Img Part Of Complex No With Same Value And The 3rd Constructor Takes 2 Argument And Initialize Real & Img Part With Diff. Values ?

Sol :-

```
#include<iostream>

using namespace std;
```

```
class ComplexNo{
    private:
        int RealPart;
        int ImPart;
    public:
        ComplexNo( ){ }
        ComplexNo(int Value){
            this->RealPart = this->ImPart=Value;
            cout<<"COMPLEX NUMBER IS ::: ("<<RealPart<<") +
                ("<<ImPart<<")i"<<endl;
        }
}
```

```

ComplexNo(int Value1 , int Value2){
    this->RealPart = Value1;
    this->ImPart = Value2;
    cout<<"COMPLEX NUMBER IS ::: ("<<RealPart<<" +
        ("<<ImPart<<" )i"<<endl;
}

void Sum(ComplexNo c1 , ComplexNo c2){
    this->RealPart = c1.RealPart + c2.RealPart;
    this->ImPart = c1.ImPart + c2.ImPart;
    cout<<"COMPLEX NUMBER IS ::: ("<<RealPart<<" +
        ("<<ImPart<<" )i"<<endl;
}

};

int main(){
    ComplexNo c1(3) , c2(5,-1) , c3;
    c3.Sum(c1,c2);
    return 0;
}

```

OUTPUT : -

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

PS E:\Yash Pandey> cd "e:\Yash Pandey\YashPandey_Codes(DSA)\C++_Codes\OOPs-Labwork_Codes\" ; g++ 3.cpp
COMPLEX NUMBER IS ::: (3) + (3)i
COMPLEX NUMBER IS ::: (5) + (-1)i
COMPLEX NUMBER IS ::: (8) + (2)i

PS E:\Yash Pandey\YashPandey_Codes(DSA)\C++_Codes\OOPs-Labwork_Codes>
```

PROGRAM-10.

Q :- WAP To Find The Greatest Of 2 Given Number In 2 Different Classes Using Friend Function ?

Sol :-

```
#include<iostream>
```

```
using namespace std;
```

```
class SecondNum;    //Forward Declaration Of Class 'SecondNum'.....
```

```
class FirstNum{
```

```
    private:    int Num1;
```

```
    public:
```

```
    FirstNum(int a) { this->Num1=a;    }
```

```
    friend void Compare(FirstNum,SecondNum);
```

```
};
```

```
class SecondNum{
```

```
    private:    int Num2;
```

```
    public:
```

```
    SecondNum(int b) {          this->Num2=b;    }
```

```
    friend void Compare(FirstNum,SecondNum);
```

```
};
```

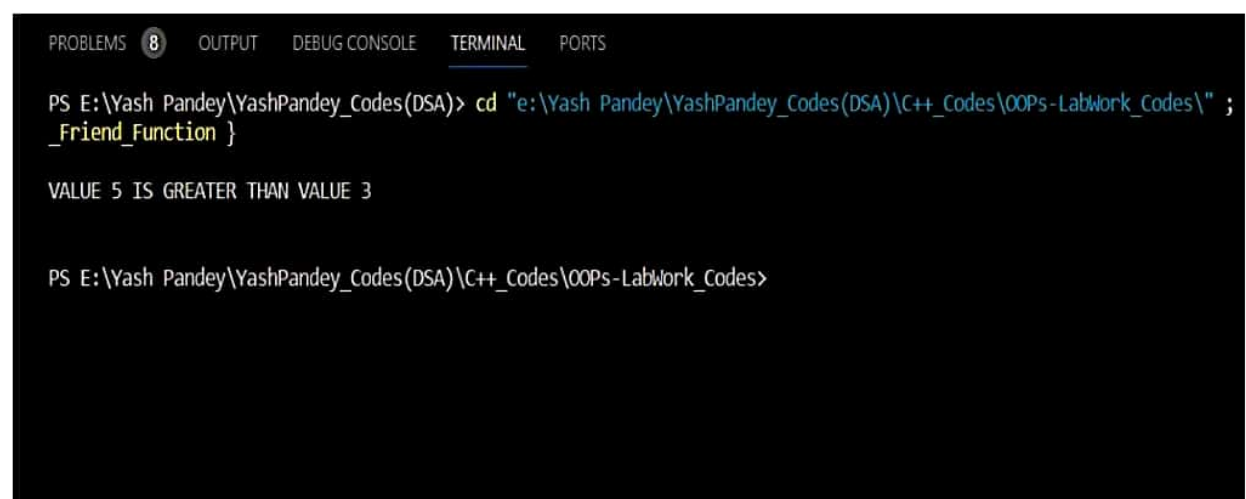
```

void Compare(FirstNum n1 , SecondNum n2){
    int Num1=n1.Num1 , Num2=n2.Num2;
    if(Num1>=Num2){
        cout<<"VALUE "<<Num1<<" IS GREATER THAN VALUE "<<Num2<<endl;
    } else{
        cout<<"VALUE "<<Num2<<" IS GREATER THAN VALUE "<<Num1<<endl;
    }
}

int main(){
    FirstNum n1(3); SecondNum n2(5);
    Compare(n1,n2);
    return 0;
}

```

OUTPUT : -



```

PROBLEMS 8 OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS E:\Yash Pandey\YashPandey_Codes(DSA)> cd "e:\Yash Pandey\YashPandey_Codes(DSA)\C++_Codes\OOPs-Labwork_Codes\" ;
_Friend_Function }
VALUE 5 IS GREATER THAN VALUE 3
PS E:\Yash Pandey\YashPandey_Codes(DSA)\C++_Codes\OOPs-Labwork_Codes>

```

PROGRAM-11.

Q :- WAP To Find The Biggest Of 3 Numbers 3 Different Classes
With Help Of Friend Function ?

Sol :-

```
#include<iostream>

using namespace std;

class SecondNum;      //Forward Declaration Of Class 'SecondNum'.....
class ThirdNum;       //Forward Declaration Of Class 'ThirdNum'.....

class FirstNum{
    private:    int Num1;
    public:     FirstNum(int a) { this->Num1=a;    }

                friend void Compare(FirstNum , SecondNum , ThirdNum);
};

class SecondNum{
    private:    int Num2;
    public:     SecondNum(int b){    this->Num2=b;    }

                friend void Compare(FirstNum , SecondNum , ThirdNum);
};

class ThirdNum{
    private:    int Num3;
```

```

        public:      ThirdNum(int c){  this->Num3=c;    }

        friend void Compare(FirstNum , SecondNum , ThirdNum);

};

void Compare(FirstNum n1,SecondNum n2,ThirdNum n3){

    int Num1=n1.Num1 , Num2=n2.Num2 , Num3=n3.Num3;

    int ans= (Num1>Num2) ?

        ( (Num1>Num3) ? Num1 : Num3 ) : ( (Num2>Num3) ? Num2 : Num3 );

    cout<<"THE GREATEST VALUE BETWEEN THE VALUES "<<Num1<<" ,

        "<<Num2<<" & "<<Num3<<" IS ::: "<<ans<<endl;

}

int main(){

    FirstNum n1(3);      SecondNum n2(7);

    ThirdNum n3(5);      Compare(n1,n2,n3);

    return 0;

}

```

OUTPUT : -



```

PROBLEMS 8 OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS E:\Yash Pandey\YashPandey_Codes(DSA)> cd "e:\Yash Pandey\YashPandey_Codes(DSA)\C++_Codes\OOPs-LabWork_Codes\" ;
d_Class }

THE GREATEST VALUE BETWEEN THE VALUES 3 , 7 & 5 IS ::: 7

PS E:\Yash Pandey\YashPandey_Codes(DSA)\C++_Codes\OOPs-LabWork_Codes>

```


PROGRAM-12.

Q :- WAP To Find The Sum Of 2 Numbers Declared In A Class And Display The Numbers & Sum Using Friend Function ?

Sol :-

```
#include<iostream>
```

```
using namespace std;
```

```
class Display;    //Forward Declaration Of Class 'Display'.....
```

```
class SetNum{
```

```
    private:
```

```
        int Num1 , Num2;
```

```
    public:
```

```
        SetNum(int a,int b){
```

```
            this->Num1=a;    this->Num2=b;
```

```
        }
```

```
        friend class Display;
```

```
};
```

```
class Display{
```

```
    private:    int sum;
```

```
    public:
```

```

    Display( SetNum& n){

        sum=n.Num1+n.Num2;

        cout<<"THE 1st NUMBER IS ::: "<<n.Num1<<endl;

        cout<<"THE 2st NUMBER IS ::: "<<n.Num2<<endl;

        cout<<"SUM OF "<<n.Num1<<" AND "<<n.Num2<<" IS ::: " <sum<<endl;

    }

};

int main(){

    SetNum n (3,7);

    Display disp (n);

    return 0;

}

```

OUTPUT : -



```

PROBLEMS 8 OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS E:\Yash Pandey\YashPandey_Codes(DSA)> cd "e:\Yash Pandey\YashPandey_Codes(DSA)\C++_Codes\OOPS-LabWork_Codes\" ;
\OOPS10_FriendClass_Basic }

THE 1st NUMBER IS ::: 3
THE 2st NUMBER IS ::: 7
SUM OF 3 AND 7 IS ::: 10

PS E:\Yash Pandey\YashPandey_Codes(DSA)\C++_Codes\OOPS-LabWork_Codes>

```

PROGRAM-13.

Q :- WAP To Implement Class String Using Operator Overloading :

- 1. Overload(+) Operator To Carry Concatenation Of String .**
- 2. Overload(==) Operator To Perform Copy Of String..**
- 3. Overload(+) Operator To Perform Comparisn Of String .**

Sol :-

```
#include<iostream>
```

```
#include<string.h>
```

```
using namespace std;
```

```
class String{
```

```
    public:
```

```
        char* ptr;  int Length;
```

```
        String( ){    Length=0; ptr=NULL;    }
```

```
        String(string str){
```

```
            char* s = &str[0];  Length = strlen(s);
```

```
            ptr=new char[Length+1];    strcpy(ptr , s);
```

```
        }
```

```
        String( String &s1 ){
```

```
            Length = s1.Length;        ptr = new char[Length+1];
```

```
            strcpy(ptr,s1.ptr);
```

```
        }
```

```

String operator+ (String &str1){
    String temp;      temp.Length = Length + str1.Length;
    temp.ptr=new char[temp.Length +1];
    strcpy(temp.ptr , ptr);   strcat(temp.ptr,str1.ptr);
    return temp;
}

void operator== (String &str1){
    Length=str1.Length;    ptr=new char[Length+1];
    strcpy(ptr,str1.ptr);
}

int operator<= (String &s1){
    int First_str=Length , Second_str=s1.Length;
    return (First_str <= Second_str);
}

friend void Display(String);
};

void Display(String str){
    cout<<"THE STRING IS : : : "<<str.ptr<<endl;
}

int main(){
    String str1("Yash_");   String str2("Pandey");
    Display(str1);          Display(str2);
    String str3;            str3=str1+str2;

```

```

        cout<<"After operater+ () Function , ";      Display(str3);

String str4;      str4==str3;

cout<<"COPYING DATA OF str3->str4 , ";      Display(str4);

int compare=str1<=str2;

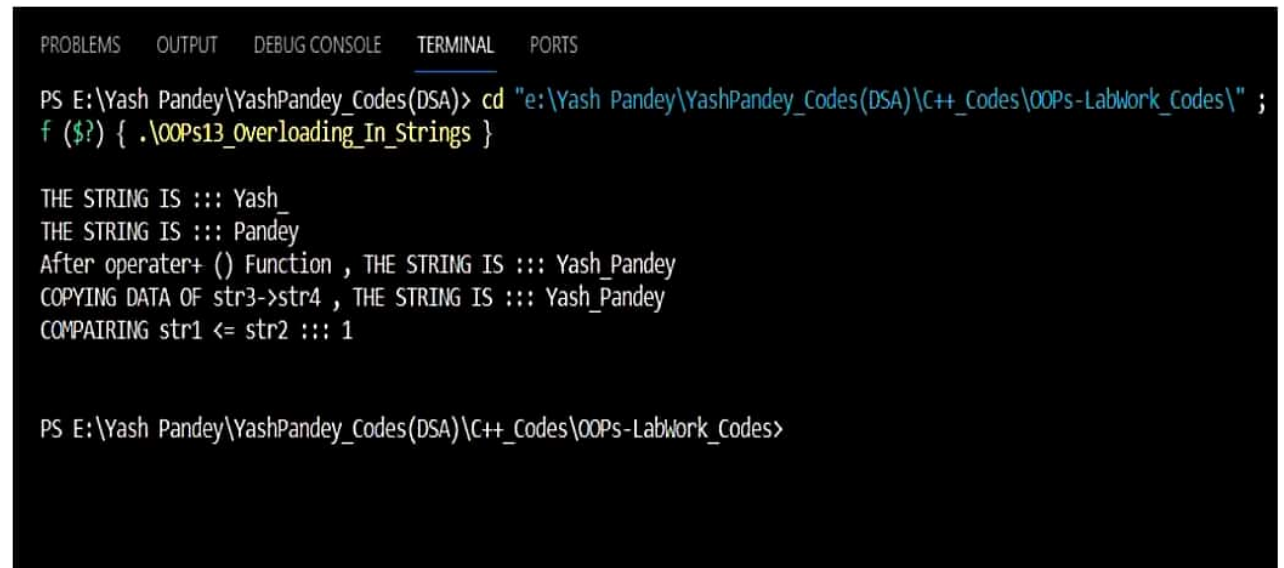
cout<<"COMPAIRING str1 <= str2 ::: "<<compare<<endl;

return 0;

}

```

OUTPUT : -



The screenshot shows a terminal window with the following content:

```

PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

PS E:\Yash Pandey\YashPandey_Codes(DSA)> cd "e:\Yash Pandey\YashPandey_Codes(DSA)\C++_Codes\OOPs-LabWork_Codes\" ;
f ($?) { .\OOPs13_Overloading_In_Strings }

THE STRING IS ::: Yash_
THE STRING IS ::: Pandey
After operater+ () Function , THE STRING IS ::: Yash_Pandey
COPYING DATA OF str3->str4 , THE STRING IS ::: Yash_Pandey
COMPAIRING str1 <= str2 ::: 1

PS E:\Yash Pandey\YashPandey_Codes(DSA)\C++_Codes\OOPs-LabWork_Codes>

```

PROGRAM-14.

Q :- WAP To Implement Below Parts :-

- 1. Function To Find The Length Of String..**
- 2. Function To Convert Lowercase To Uppercase In String..**
- 3. Function To Convert Uppercase To Lowercase In String..**

Sol :-

```
#include<iostream>
```

```
using namespace std;
```

```
class StringOperation{
```

```
    public:
```

```
        StringOperation( ){ }
```

```
        int GetLength(char str1[]){
```

```
            char* ptr=str1;    int Len = 0;
```

```
            while( ptr[Len] ){ Len++;    }
```

```
            return Len;
```

```
        }
```

```
        void To_Uppercase(char str1[] ){
```

```
            int Size=GetLength(str1);
```

```
            for(int i = 0 ; i < Size ; i++){
```

```
                int Val=(int) str1[i];
```

```

        if(Val>=97 && Val<=122){str1[i] =str1[i] - ' ';    }
    }
    cout<<"THE Uppercase-STRING IS ::: ";
    for(int i = 0 ; i<Size ; i++){        cout<<str1[i]<<"";        }
}

void To_Lowercase(char str1[]){
    int Size=GetLength(str1);
    for(int i=0 ; i<Size ; i++){
        int Val=(int) str1[i];
        if(Val>=65 && Val<=90){            str1[i] =str1[i]+' ';            }
    }
    cout<<"THE Lowercase-STRING IS ::: ";
    for(int i=0 ; i<Size ; i++){        cout<<str1[i]<<"";        }
}

};

int main(){
    char ch[]="Yash_Pandey";
    StringOperation str1;
    int Len=str1.GetLength(ch);
    cout<<"THE LENGTH OF STRING '"<<ch<<"' IS ::: "<<Len<< endl;
    str1.To_Uppercase(ch);
    str1.To_Lowercase(ch);
    return 0;
}

```

OUTPUT : -

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS E:\Yash Pandey\YashPandey_Codes(DSA)> cd "e:\Yash Pandey\YashPandey_Codes(DSA)\C++_Codes\OOPs-LabWork_Codes\" ;
```

```
THE LENGTH OF STRING 'Yash_Pandey' IS ::: 11
```

```
THE Uppercase-STRING IS ::: YASH_PANDEY
```

```
THE Lowercase-STRING IS ::: yash_pandey
```

```
PS E:\Yash Pandey\YashPandey_Codes(DSA)\C++_Codes\OOPs-LabWork_Codes>
```


PROGRAM-15.

Q :- WAP To Overload Unary Increment(++) Operator ?

Sol :-

```
#include<iostream>

using namespace std;

class Overload{

    public:    int Num;

              void operator++ ( ) {    this->Num++;    }

              void Display(){    cout<<"THE VALUE IS ::: "<<Num<<endl;    }

};

int main(){

    Overload obj;    obj.Num=10;        obj.Display();

    ++obj;        obj.Display();        return 0;

}
```

OUTPUT : -



```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

PS E:\Yash Pandey\YashPandey_Codes(DSA)> cd "e:\Yash Pandey\YashPandey_Codes(DSA)\C++_Codes\OOPS-LabWork_Codes\OOPS_04_Operator_OverLoading\" ;
Ps15_Unary_Operator_OverLoading } ; if ($?) { .\OOPS15_Unary_Operator_OverLoading }

THE VALUE IS ::: 10
THE VALUE IS ::: 11

PS E:\Yash Pandey\YashPandey_Codes(DSA)\C++_Codes\OOPS-LabWork_Codes\OOPS_04_Operator_OverLoading>
```

PROGRAM-16.

Q :- WAP To Overload new And delete Operator ?

Sol :-

```
#include<iostream>
```

```
using namespace std;
```

```
class Student {
```

```
    private:
```

```
        string name;
```

```
        int age;
```

```
    public:
```

```
        Student( ) { }
```

```
        Student(string name, int age) {
```

```
            this->name = name;
```

```
            this->age = age;
```

```
        }
```

```
        void display() {
```

```
            cout<< "NAME ::: " << this->name << endl;
```

```
            cout<< "AGE ::: " << this->age << endl;
```

```
        }
```

```

void * operator new(size_t size){

    cout<< "Overloading new operator with size ::: " << size << endl;

    void * p = ::operator new(size);

    return p;

}

void operator delete(void * p) {

    cout<< "Overloading delete operator..... " << endl;

    free(p);

}

};

int main() {

    Student* p = new Student("Yash", 122);

    p->display();  delete p;

    return 0;

}

```

OUTPUT : -



```

PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

PS E:\Yash Pandey\YashPandey_Codes(DSA)> cd "e:\Yash Pandey\YashPandey_Codes(DSA)\C++_Codes\OOPs-LabWork_Codes\OOPs_04_Operater_OverLoading\" ;
Ps16_New_And_Delete_Overloading } ; if ($?) { .\OOPs16_New_And_Delete_Overloading }

Overloading new operator with size ::: 28
NAME ::: Yash
AGE ::: 122
Overloading delete operator.....

PS E:\Yash Pandey\YashPandey_Codes(DSA)\C++_Codes\OOPs-LabWork_Codes\OOPs_04_Operater_OverLoading>

```

PROGRAM-17.

Q :- WAP create a base class with data members like name , rollno , gender and two member functions getdata and display. Derive a class physical fitness from basic info which has data members height and weight and member functions getdata display().Display all the information using object of derived class.?

Sol :-

```
#include <iostream>
```

```
using namespace std;
```

```
class Basic_Info {
```

```
    private:
```

```
        string Name;        int RollNo;        char Gender;
```

```
    public:
```

```
        void getdata_Base(string name,int rollno,char gender) {
```

```
            this->Name=name;        this->RollNo=rollno;
```

```
            this->Gender=gender;
```

```
        }
```

```
        void Display_Base(){
```

```
            cout<<"NAME OF THE STUDENT IS ::: "<<Name<<endl;
```

```
            cout<<"ROLL No OF THE STUDENT IS ::: "<<RollNo<<endl;
```

```
            cout<<"GENDER OF THE STUDENT IS ::: "<<Gender<<endl;
```

```
        }
```

```
};
```

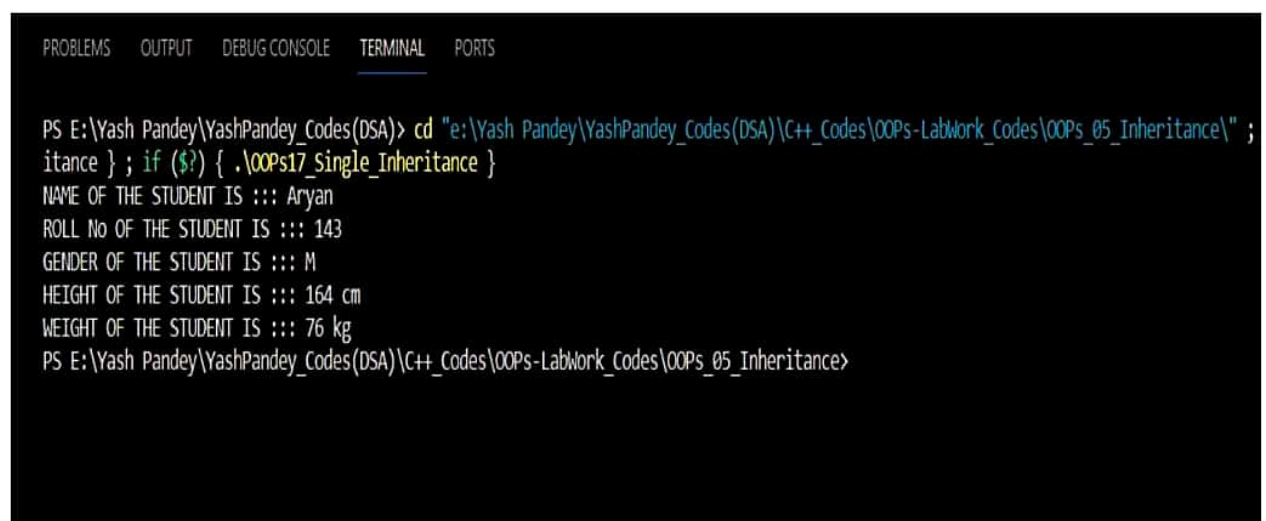
```

class Physical_Fit : public Basic_Info {
    private:    int Height;  int Weight;
    public:
        void getdata_Derived(int h,int w) {
            this->Height=h;    this->Weight=w;
        }
        void Display_Derived(){
            cout<<"HEIGHT OF THE STUDENT IS ::: "<<Height<<" cm"<<endl;
            cout<<"WEIGHT OF THE STUDENT IS ::: "<<Weight<<" kg"<<endl;
        }
};

int main() {
    Physical_Fit derived;
    derived.getdata_Base("Aryan",143,'M');
    derived.getdata_Derived(164,76);  derived.Display_Base();
    derived.Display_Derived();        return 0;
}

```

OUTPUT : -



The screenshot shows a terminal window with the following content:

```

PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

PS E:\Yash Pandey\YashPandey_Codes(DSA)> cd "e:\Yash Pandey\YashPandey_Codes(DSA)\C++_Codes\OOPS-LabWork_Codes\OOPS_05_Inheritance\" ;
itance } ; if ($?) { .\OOPS17_Single_Inheritance }
NAME OF THE STUDENT IS ::: Aryan
ROLL No OF THE STUDENT IS ::: 143
GENDER OF THE STUDENT IS ::: M
HEIGHT OF THE STUDENT IS ::: 164 cm
WEIGHT OF THE STUDENT IS ::: 76 kg
PS E:\Yash Pandey\YashPandey_Codes(DSA)\C++_Codes\OOPS-LabWork_Codes\OOPS_05_Inheritance>

```

PROGRAM-18.

Q :- WAP To create a base class with data members as Book No And Book Name With Member Functions As GetData And Display . Create A Second Class With Data Members As Author Name And Publisher Name And Member Function As GetData And SetData . Derive The Class Third From The First Class And Second Class With Data Members As No Of Pages And Year Of Publication With Member Functions As GetData And Display . Display All The Information Using Array Of Objects Of The Third Class ?

Sol :-

```
#include<iostream>

using namespace std;

class First{
    private:  int Book_No;      string Book_Name;
    public:

        void GetData1(int num,string name){
            this->Book_No=num;  this->Book_Name=name;
        }

        void Display1(){
            cout<<"BOOK NUMBER IS ::: "<<Book_No<<" And BOOK NAME IS ::: 
            "<<Book_Name<<endl;
        }
};
```

```

class Second{
    private:   string Author_Name , Publishor_Name;
    public:
        void GetData2(string author,string pub){
            this->Author_Name=author;   this->Publishor_Name=pub;
        }
        void Display2(){
            cout<<"AUTHOR NAME IS ::: "<<Author_Name<<" And
                PUBLISHOR NAME IS ::: "<<Publishor_Name<<endl;
        }
};

class Third : public First , public Second{
    private:   int Pages , Year_Of_Pub;
    public:
        void GetData3(int num1,int num2){
            this->Pages=num1;           this->Year_Of_Pub=num2;
        }
        void Display3(){
            cout<<"NUMBER OF PAGES ARE ::: "<<Pages<<" And YEAR OF
                PUBLICATION IS ::: "<<Year_Of_Pub<<endl;
        }
};

```

```

int main(){

    Third obj[2];

    obj[0].GetData1(101,"OOPs Using C++");

    obj[0].GetData2("E Balaguruswamy","McGraw Hill");

    obj[0].GetData3(700,2010);

    obj[0].Display1();          obj[0].Display2();          obj[0].Display3();

    obj[1].GetData1(102,"Data Structures In C");

    obj[1].GetData2("Schawm","McGraw Hill");

    obj[1].GetData3(800,2008);

    obj[1].Display1();          obj[1].Display2();          obj[1].Display3();

    return 0;

}

```

OUTPUT : -

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```

PS E:\Yash Pandey\YashPandey_Codes(DSA)> cd "e:\Yash Pandey\YashPandey_Codes(DSA)\C++_Codes\OOPs-LabWork_Codes\OOPs_05_Inheritance\" ;
nheritance } ; if ($?) { .\OOPs18_Multiple_Inheritance }

```

```

BOOK NUMBER IS ::: 101 And BOOK NAME IS ::: OOPs Using C++
AUTHOR NAME IS ::: 'E Balaguruswamy' And PUBLISHER NAME IS ::: McGraw Hill
NUMBER OF PAGES ARE ::: 700 And YEAR OF PUBLICATION IS ::: 2010

```

```

BOOK NUMBER IS ::: 102 And BOOK NAME IS ::: Data Structures In C
AUTHOR NAME IS ::: 'Schawm' And PUBLISHER NAME IS ::: McGraw Hill
NUMBER OF PAGES ARE ::: 800 And YEAR OF PUBLICATION IS ::: 2008

```

```

PS E:\Yash Pandey\YashPandey_Codes(DSA)\C++_Codes\OOPs-LabWork_Codes\OOPs_05_Inheritance>

```


PROGRAM-19.

Q :- WAP To Design 3 classes Student Exam and Result . The student class has data members such as roll number , name . Create a class exam by Inheriting From the student class . The exam class adds data member representing the marks Scored in 6 subjects . derive the result from the exam class and it has its own data members such as total marks . WAP to model this relationship?

Sol :-

```
#include<iostream>

using namespace std;

class Student{

    private:

        int Roll_No;        string Name;

    public:

        void GetData1(int num,string name){

            this->Roll_No=num;    this->Name=name;

        }

        void Display1(){

            cout<<"Roll Number Of "<<Name<<" IS ::: "<<Roll_No<<endl;

        }

};
```

```

class Exam : public Student{

    private:

        int Marks[6]; int TotalMarks=0;

    public:

        void GetData2(int n , int n1 , int n2 , int n3 , int n4 , int n5 ){

            this->Marks[0]=n;  this->Marks[1]=n1; this->Marks[2]=n2;

            this->Marks[3]=n3;  this->Marks[4]=n4;      this->Marks[5]=n5;

            TotalMarks += ( n+n1+n2+n3+n4+n5 );

        }

        int Total(){ return TotalMarks; }

};

class Result : public Exam{

    private: float TotalMarks=0;

    public: void GetData3(){      TotalMarks=Total();      }

        void Display3(){ cout<<"PERCENTAGE IN 6 SUBJECTS IS :::

            "<<(TotalMarks/6)<<"%"<<endl;

        }

};

int main(){

    Result res;

    res.GetData1(122,"Yash Pandey");

    res.GetData2(89,90,93,96,91,93);

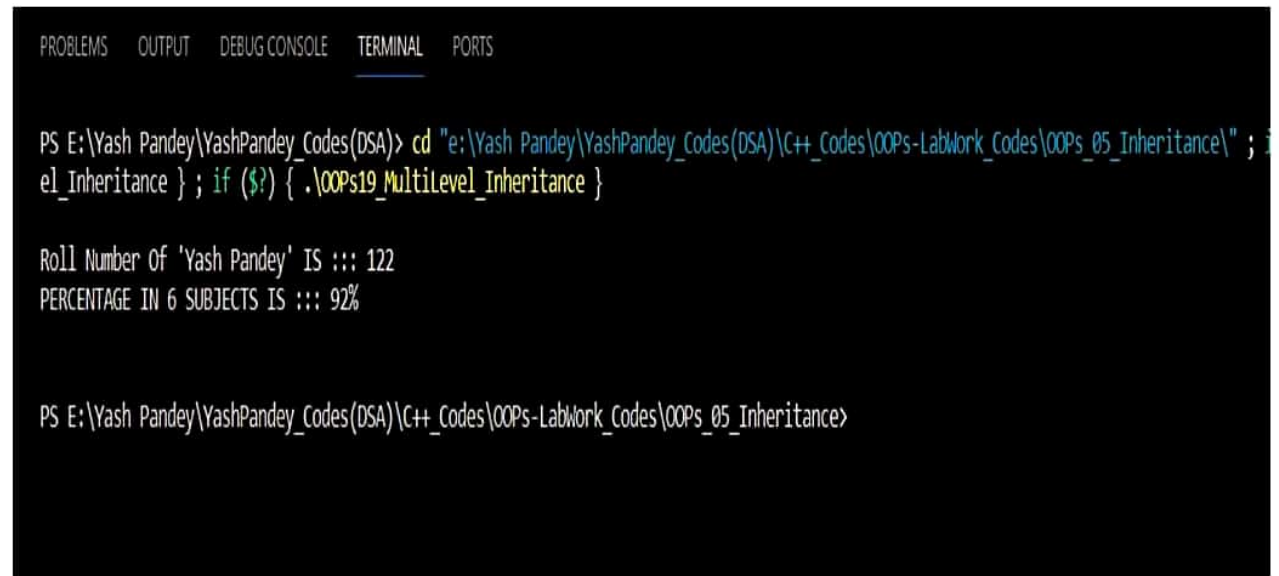
    res.GetData3();

    res.Display1();

```

```
res.Display3();  
  
return 0;  
  
}
```

OUTPUT : -



```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS  
  
PS E:\Yash Pandey\YashPandey_Codes(DSA)> cd "e:\Yash Pandey\YashPandey_Codes(DSA)\C++_Codes\OOPs-LabWork_Codes\OOPs_05_Inheritance\" ; if ($?) { .\OOPs19_MultiLevel_Inheritance }  
  
Roll Number Of 'Yash Pandey' IS ::: 122  
PERCENTAGE IN 6 SUBJECTS IS ::: 92%  
  
PS E:\Yash Pandey\YashPandey_Codes(DSA)\C++_Codes\OOPs-LabWork_Codes\OOPs_05_Inheritance>
```

PROGRAM-20.

Q :- WAP To Create a base class called Shape . Use this class to store 2 Data values . Derive two specific classes called Triangle and Rectangle from the base class & Add it to the base class , a Member function getdata to initialize this class data members and another Member function display to compute and display the area of the figures . Make display a virtual function and redefine this function in the derived classes to suit the requirements . Using these three classes design a program that will accept driven of a Triangle or Rectangle interactively and display the area?

Sol :-

```
#include<iostream>

using namespace std;

class Shape{
    public:    int Dim1 , Dim2;

    void GetData(){
        cout<<"ENTER THE VALUE OF VARIABLE Dim1 ::: ";
        cin>>Dim1;

        cout<<"ENTER THE VALUE OF VARIABLE Dim2 ::: ";
        cin>>Dim2;

    }
```

```

        virtual void Display(){
            cout<<"AREA OF FIGURE IS ::: "<<(Dim1*Dim2)<<endl;
        }
};

class Triangle : public Shape{
public:
    void Display(){
        cout<<"USING Display() Function Of Class Triangle....."<<endl;
        cout<<"THE AREA OF TRIANGLE IS ::: "<<(0.5*Dim1*Dim2)<<endl;
    }
};

class Rectangle : public Shape{
public:
    void Display(){
        cout<<"USING Display() Function Of Class Rectangle....."<<endl;
        cout<<"THE AREA OF RECTANGLE IS ::: "<<(Dim1*Dim2)<<endl;
    }
};

int main(){
    Shape* Base_ptr;
    int Select;
    cout<<"ENTER '1' FOR Triangle Or '2' FOR Rectangle ::: ";
    cin>>Select;

```

```

if(Select==1){
    Base_ptr=new Triangle;
    Base_ptr->GetData();
    Base_ptr->Display();
} else if(Select==2){
    Base_ptr=new Rectangle;
    Base_ptr->GetData();
    Base_ptr->Display();
} else{
    cout<<"ENTERED WRONG SELECTION....."<<endl;
}
return 0;
}

```

OUTPUT : -

```

PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

PS E:\Yash Pandey\YashPandey_Codes(DSA)> cd "e:\Yash Pandey\YashPandey_Codes(DSA)\C++_Codes\OOPS-LabWork_Codes\OOPS_06_Polymorphism\" ;
n } ; if ($?) { .\OOPS23_VirtualFunction }

ENTER '1' FOR Triangle Or '2' FOR Rectangle ::: 1
ENTER THE VALUE OF VARIABLE Dim1 ::: 10
ENTER THE VALUE OF VARIABLE Dim2 ::: 20

USING Display() Function Of Class Triangle.....
THE AREA OF TRIANGLE IS ::: 100

PS E:\Yash Pandey\YashPandey_Codes(DSA)\C++_Codes\OOPS-LabWork_Codes\OOPS_06_Polymorphism>

```

PROGRAM-21.

Q :- WAP To Create a class called LIST with two pure virtual function store() and retrieve() To store a value call store and to retrieve call retrieve function. Derive two classes stack and queue from it and override store and retrieve?

Sol :-

```
#include<iostream>

using namespace std;

class Node{
    public:      int data;      Node* next;

    Node(int data){
        this->data=data;  this->next=NULL;
    }
};

class List{
    public:

        Node* Head , Tail; int Size=0;

        List(){ }      // Default Constructor Of Class List.....

        virtual void Store(int data)=0;  // Pure Virtual Function.....

        virtual void Retrieve()=0;      // Pure Virtual Function.....
};
```

```

class Stacks : public List{

    private:      Node* Top;

    public:

        Stacks(){

            this->Head=this->Tail=this->Top=NULL;

        }

        void Store (int data) override{

            Node* NewNode=new Node(data);

            if(Head==NULL){

                Head=Tail=Top=NewNode;

                Size++;      return;

            }

            NewNode->next=Head;

            Head=Top=NewNode;   Size++;

        }

        void Retrieve() override{

            cout<<"THE STACK IS ::: (Top)--> ";

            Node* Temp=Top;

            while(Temp->next!=NULL){

                cout<<Temp->data<<" , ";      Temp=Temp->next;

            }

            cout<<Temp->data<<endl;

            cout<<"TOTAL No OF ELEMENTS IN THE STACK ARE ::: "<<Size<<endl;

            cout<<endl;

```



```

    }

};

class Queues : public List{

private:   Node* Front , Rear;

public:

    Queues(){

        this->Head=this->Tail=this->Front=this->Rear=NULL;

    }

    void Store(int data) override{

        Node* NewNode=new Node(data);

        if(Head==NULL){

            Head=Tail=Front=Rear=NewNode;

            Size++;      return;

        }

        Tail->next=NewNode;

        Tail=Rear=NewNode;    Size++;

    }

    void Retrieve() override{

        cout<<"THE QUEUE IS ::: (Front) --> ";

        Node* temp=Front;

        while(temp->next!=NULL){

            cout<<temp->data<<" , ";

            temp=temp->next;

        }    cout<<temp->data<<" <-- (Rear)"<<endl;

```

```

        cout<<"TOTAL No OF ELEMENTS IN THE QUEUE ARE ::: "<<Size<<endl;

        cout<<endl;

    }

};

int main(){

    Stacks s;    Queues q;

    for(int i = 1 ; i <= 5 ; i++){

        s.Store(i*10);    q.Store(i*10);

    }

    s.Retrieve();    q.Retrieve();

    return 0;

}

```

OUTPUT : -

```

PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

PS E:\Yash Pandey\YashPandey_Codes(DSA)> cd "e:\Yash Pandey\YashPandey_Codes(DSA)\C++_Codes\OOPS-LabWork_Codes\OOPS_06_Polymorphism\" ;
alFunction_Question } ; if ($?) { .\OOPS29_VirtualFunction_Question }

THE STACK IS ::: (Top)--> 50 , 40 , 30 , 20 , 10
TOTAL No OF ELEMENTS IN THE STACK ARE ::: 5

THE QUEUE IS ::: (Front) --> 10 , 20 , 30 , 40 , 50 <-- (Rear)
TOTAL No OF ELEMENTS IN THE QUEUE ARE ::: 5

PS E:\Yash Pandey\YashPandey_Codes(DSA)\C++_Codes\OOPS-LabWork_Codes\OOPS_06_Polymorphism>

```

PROGRAM-22.

Q :- WAP To Define A Function Template For Calculating The Square Of Given Numbers With Different Data Types ?

Sol :-

```
#include<iostream>

using namespace std;

template<class T>

void Square(T Val){

    cout<<"SQUARE OF VALUE "<<Val<<" IS ::: "<<(Val*Val)<<endl;

}

int main(){

    Square(int(25));        Square(long(16));

    Square(float(6.73));    Square(double(7.8942));

    return 0;

}
```

OUTPUT : -



```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS  SEARCH TERMINAL OUTPUT  COMMENTS

PS E:\Yash Pandey\YashPandey_Codes(DSA)> cd "e:\Yash Pandey\YashPandey_Codes(DSA)\C++_Codes\OOPS-LabWork_Codes\OOPS_07_Templates\" ;
ate_Ques2 } ; if ($?) { .\OOPS31_Template_Ques2 }
SQUARE OF VALUE 25 IS ::: 625
SQUARE OF VALUE 16 IS ::: 256
SQUARE OF VALUE 6.73 IS ::: 45.2929
SQUARE OF VALUE 7.8942 IS ::: 62.3184
PS E:\Yash Pandey\YashPandey_Codes(DSA)\C++_Codes\OOPS-LabWork_Codes\OOPS_07_Templates>
```

PROGRAM-23.

Q :- WAP To demonstrate the use of special functions, constructor and destructor in the class template. The program is used to find the bigger of two entered numbers?

Sol :-

```
#include<iostream>
```

```
using namespace std;
```

```
template<class T>
```

```
class Greater{
```

```
    private:
```

```
        T Num1 , Num2;
```

```
    public:
```

```
        Greater(T Num1,T Num2){
```

```
            this->Num1=Num1;    this->Num2=Num2;
```

```
        }
```

```
        ~Greater(){
```

```
            cout<<"OBJECT WITH DATA MEMBER "<<Num1<<" &
```

```
            "<<Num2<<" IS DESTROYED...."<<endl;
```

```
        }
```

```

    T Compare(){
        return ((Num1>=Num2) ? Num1 : Num2) ;
    }
};

int main(){
    int a=5,b=4;

    float c=3.7,d=8.3;

    Greater<int> Integer(a,b);

    Greater<float> Float(c,d);

    cout<<"GREATER VALUE Between "<<a<<" & "<<b<<" IS :::

        "<<Integer.Compare()<<endl;

    cout<<"GREATER VALUE Between "<<c<<" & "<<d<<" IS :::

        "<<Float.Compare()<<endl;

    return 0;
}

```

OUTPUT : -

```

PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS  SEARCH TERMINAL OUTPUT  COMMENTS

PS E:\Yash Pandey\YashPandey_Codes(DSA)> cd "e:\Yash Pandey\YashPandey_Codes(DSA)\C++_Codes\OOPS-LabWork_Codes\OOPS_07_Templates\" ;
if ($?) { .\OOPS32_Template_Ques3 }

GREATER VALUE Between 5 & 4 IS ::: 5
GREATER VALUE Between 3.7 & 8.3 IS ::: 8.3

OBJECT WITH DATA MEMBER 3.7 & 8.3 IS DESTROYED....
OBJECT WITH DATA MEMBER 5 & 4 IS DESTROYED....
PS E:\Yash Pandey\YashPandey_Codes(DSA)\C++_Codes\OOPS-LabWork_Codes\OOPS_07_Templates>

```

PROGRAM-24.

Q :- Write a program to perform the deletion of white spaces such as horizontal tab, vertical tab, space ,line feed ,new line and carriage return from a text file and store the contents of the file without the white spaces on another file?

Sol :-

```
#include <iostream>

#include <fstream>

using namespace std;

int main(){

    ifstream Input1("sample.txt");

    char SingleChar;

    cout << "THE Input1 TEXT IS ::: "<< endl;

    while (Input1.good()){

        Input1.get(SingleChar);

        cout << SingleChar;

    }

    ifstream Input("sample.txt");

    ofstream Output("out.txt");

    Input.get(SingleChar);
```

```

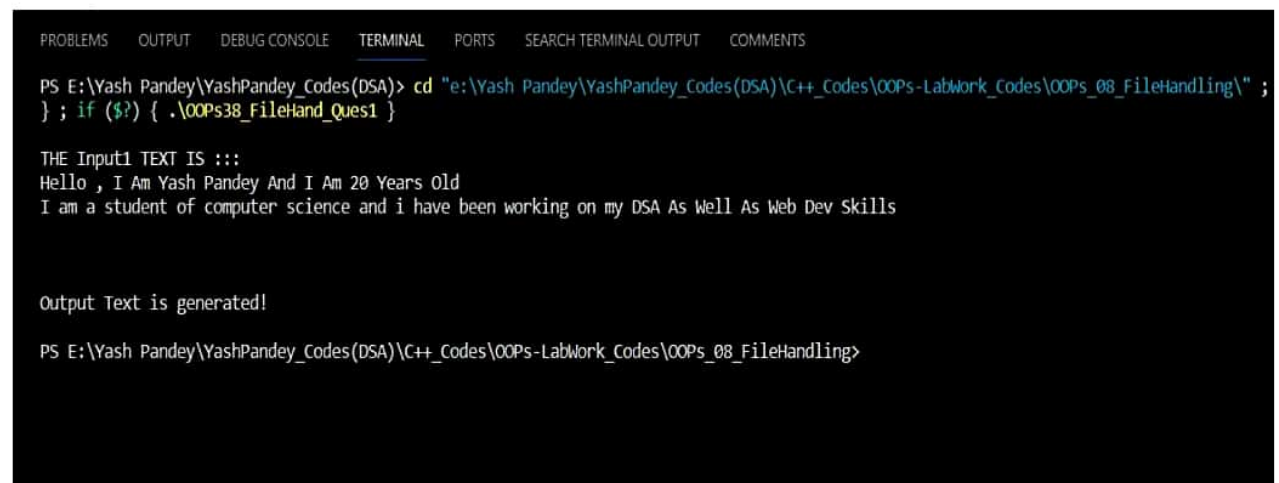
while (Input.good()){
    if ((SingleChar != ' ') && (SingleChar != '\n') &&
        (SingleChar != '\t') && (SingleChar != '\r')){
        Output << SingleChar;
    }
    Input.get(SingleChar);
}

cout << "Output Text is generated!" << endl;

return 0;
}

```

OUTPUT : -



```

PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS  SEARCH TERMINAL OUTPUT  COMMENTS

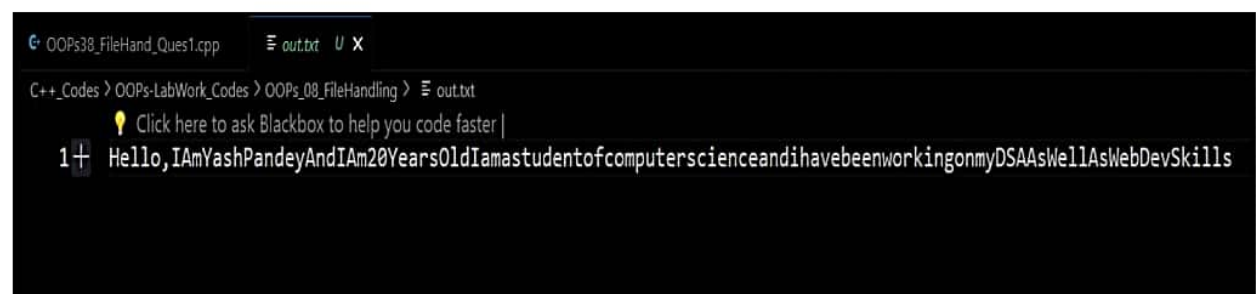
PS E:\Yash Pandey\YashPandey_Codes(DSA)> cd "e:\Yash Pandey\YashPandey_Codes(DSA)\C++_Codes\OOPS-LabWork_Codes\OOPS_08_FileHandling\" ;
} ; if ($?) { .\OOPS38_FileHand_Ques1 }

THE Input1 TEXT IS :::
Hello , I Am Yash Pandey And I Am 20 Years Old
I am a student of computer science and i have been working on my DSA As Well As Web Dev Skills

Output Text is generated!

PS E:\Yash Pandey\YashPandey_Codes(DSA)\C++_Codes\OOPS-LabWork_Codes\OOPS_08_FileHandling>

```



```

OOPS38_FileHand_Ques1.cpp  out.txt  U  X

C++_Codes > OOPS-LabWork_Codes > OOPS_08_FileHandling > out.txt
Click here to ask Blackbox to help you code faster |
1+ Hello,IAmYashPandeyAndIAM20YearsOldIamastudentofcomputerscienceandihavebeenworkingonmyDSAsWellAsWebDevSkills

```

PROGRAM-25.

Q :- Write a program to read the class object of student info such as name , age ,sex ,height and weight from the keyboard and to store them on a specified file using read() and write() functions. Again the same file is opened for reading and displaying the contents of the file on the screen ?

Sol :-

```
#include<iostream>
```

```
#include<fstream>
```

```
using namespace std;
```

```
class student{
```

```
    private:
```

```
        string data = "student.txt";
```

```
        string name;int age;        char sex;
```

```
        float height , weight;
```

```
    public:
```

```
        void write(){
```

```
            ofstream MyFile;
```

```
            MyFile.open(data,ios::app);
```

```
            cout << "ENTER THE NAME OF THE STUDENT ::: ";
```

```
            cin >> name;
```

```
            cout << "ENTER THE AGE OF THE STUDENT ::: ";
```

```
            cin >> age;
```



```

    cout << "ENTER THE GENDER OF THE STUDENT ::: ";
    cin >> sex;

    cout << "ENTER THE HEIGHT OF THE STUDENT ::: ";
    cin >> height;

    cout << "ENTER THE WEIGHT OF THE STUDENT ::: ";
    cin >> weight;

    cout << endl;

    MyFile<<name<<" "<<age<<" "<<sex<<" "<<height<<" "<<weight<<endl;

    MyFile.close();
}

void read(){
    char ch;

    ifstream MyFile(data);

    cout << "THE CONTENT IN THE FILE IS ::: ";

    while (MyFile.good()) {
        MyFile.get(ch);
        cout << ch;
    }

    MyFile.close();
}

};

```

```
int main(){

    student s1;

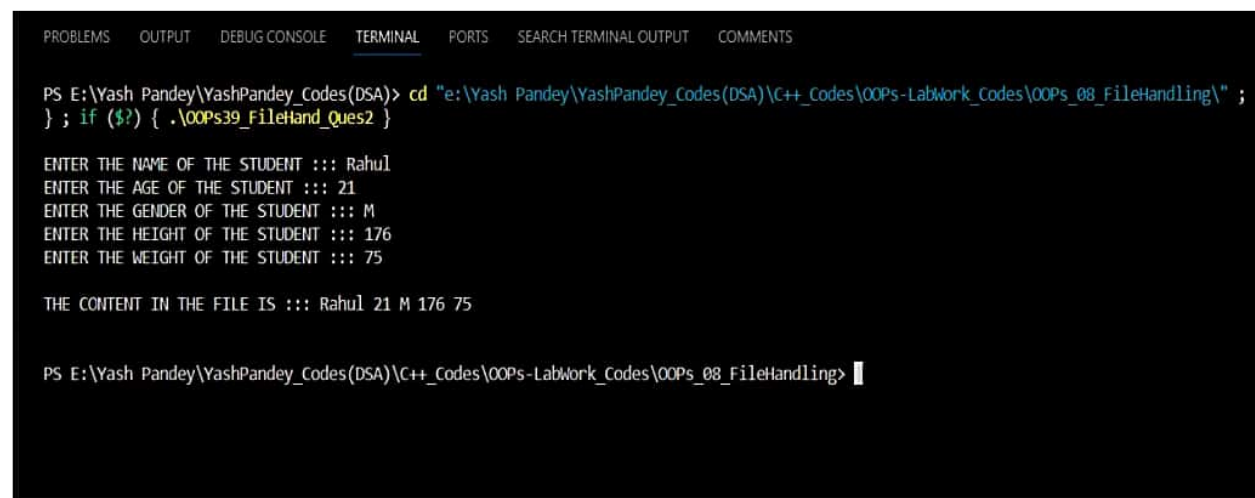
    s1.write();

    s1.read();

    return 0;

}
```

OUTPUT : -



```

PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS  SEARCH TERMINAL OUTPUT  COMMENTS

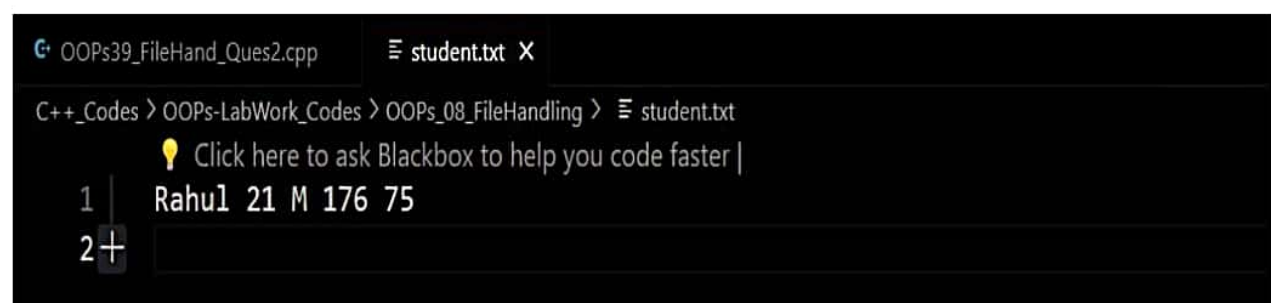
PS E:\Yash Pandey\YashPandey_Codes(DSA)> cd "e:\Yash Pandey\YashPandey_Codes(DSA)\C++_Codes\OOPs-LabWork_Codes\OOPs_08_FileHandling\" ;
} ; if ($?) { .\OOPs39_FileHand_Ques2 }

ENTER THE NAME OF THE STUDENT ::: Rahul
ENTER THE AGE OF THE STUDENT ::: 21
ENTER THE GENDER OF THE STUDENT ::: M
ENTER THE HEIGHT OF THE STUDENT ::: 176
ENTER THE WEIGHT OF THE STUDENT ::: 75

THE CONTENT IN THE FILE IS ::: Rahul 21 M 176 75

PS E:\Yash Pandey\YashPandey_Codes(DSA)\C++_Codes\OOPs-LabWork_Codes\OOPs_08_FileHandling>

```



```

OOPs39_FileHand_Ques2.cpp  student.txt X
C++_Codes > OOPs-LabWork_Codes > OOPs_08_FileHandling > student.txt
Click here to ask Blackbox to help you code faster |
1 | Rahul 21 M 176 75
2+ |

```

PROGRAM-26.

Q :- WAP To raise an exception if any attempt is made to refer to an element whose index is beyond the array size?

Sol :-

```
#include<iostream>
```

```
using namespace std;
```

```
int main(){
```

```
    int arr[6]={10,20,30,40,50,60};
```

```
    int size=sizeof(arr)/sizeof(int);
```

```
    try{
```

```
        int idx;
```

```
        cout<<"ENTER THE INDEX OF ARRAY FOR RANDOM ACCESS ::: ";
```

```
        cin>>idx;
```

```
        if(idx>=0 && idx<size){
```

```
            cout<<"VALUE AT arr["<<idx<<"] IS ::: "<<arr[idx]<<endl;
```

```
        }
```

```
        else{            throw(idx);            }
```

```
    }
```

```

        catch(int idx){

            cout<<"EXCEPTION CASE ::: INDEX "<<idx<<" IS BEYOND THE

                ARRAY SIZE.....";

        }

        return 0;

    }
}

```

OUTPUT : -

```

PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS  SEARCH TERMINAL OUTPUT  COMMENTS
PS E:\Yash Pandey\YashPandey_Codes(DSA)> cd "e:\Yash Pandey\YashPandey_Codes(DSA)\C++_Codes\OOPS-LabWork_Codes\OOPS_09_ExceptionHandling\" ;
($?) { .\OOPS47_EH_Ques3 }

ENTER THE INDEX OF ARRAY FOR RANDOM ACCESS ::: 4
VALUE AT arr[4] IS ::: 50

PS E:\Yash Pandey\YashPandey_Codes(DSA)\C++_Codes\OOPS-LabWork_Codes\OOPS_09_ExceptionHandling> cd "e:\Yash Pandey\YashPandey_Codes(DSA)\C++
?" { g++ OOPS47_EH_Ques3.cpp -o OOPS47_EH_Ques3 } ; if ($?) { .\OOPS47_EH_Ques3 }

ENTER THE INDEX OF ARRAY FOR RANDOM ACCESS ::: 9
EXCEPTION CASE ::: INDEX 9 IS BEYOND THE ARRAY SIZE.....

PS E:\Yash Pandey\YashPandey_Codes(DSA)\C++_Codes\OOPS-LabWork_Codes\OOPS_09_ExceptionHandling>

```